

Wireshark

- Graphical User Interface (GUI) with command-line option
- User-friendly with visual representation of packets
- Detailed packet analysis with protocol decoding and hierarchical display
- Advanced display and capture filters using a rich syntax
- Extensive support for various protocols across all OSI layers
- Best for in-depth analysis and troubleshooting with visual tools

Similarities

Both are network packet analyzers used to capture and analyze network traffic.

Both tools are open-source and freely available.

Support for capture filters to isolate specific traffic.

tcpdump

- Command-Line Interface (CLI) only
- Requires familiarity with command-line syntax
- Text-based output showing packet headers and summaries
- Highly scriptable; ideal for automated tasks and remote captures
- Easily captures remote traffic via SSH or other remote methods
- Primarily focuses on TCP/IP protocols